1. Strong rho, round 3: the walk step (2026-10-16)

Study: `research/rho_step3_20261016/` · **Protocol:** `PROTOCOL.md` (declared before the changes; one addendum, recorded before any timing) · **Baseline:** `ed282c5b9` (round 2) · **Host:** one x86-64 cloud VM (Intel Xeon @ 2.10 GHz, 4 vCPU), no hardware PMU.

The strong rho is the reference every Koblitz `vs_rho` comparison prices index calculus against. This round makes its walk step cheaper without changing any walk. Every output line at n = 41, 53 and 61 is identical to round 2's, timing aside, and all 14 strong-rho unit tests pass with debug assertions on.

1.1. Changes

1. **Target-feature walk.** `solve` runs its body inside a `pclmulqdq` target-feature method when the field is a folding `Gf2` on x86-64, so field products inline into the step instead of being calls to `mul_fold`. The declared closure form did not inline and was replaced (addendum).
2. **Batched walk starts.** The starts $c + js$ are formed `lanes` at a time with one batched inversion, instead of a full Itoh–Tsujii inversion (about 640 instructions) per walk. Same points, same order, same charged additions.

1.2. Instructions (measured, Callgrind, n = 53, 4 fixtures)

arm	instructions	per walk step
round 2 (`ed282c5b9`)	2,369,297,848	781
change 1	2,323,045,147	766
changes 1+2	**2,178,454,252**	**718**

The final build runs 8.1% fewer instructions per step (1.09×) than round 2, and 23.5% fewer than round 1's baseline of 938. Output digests are identical in every arm.

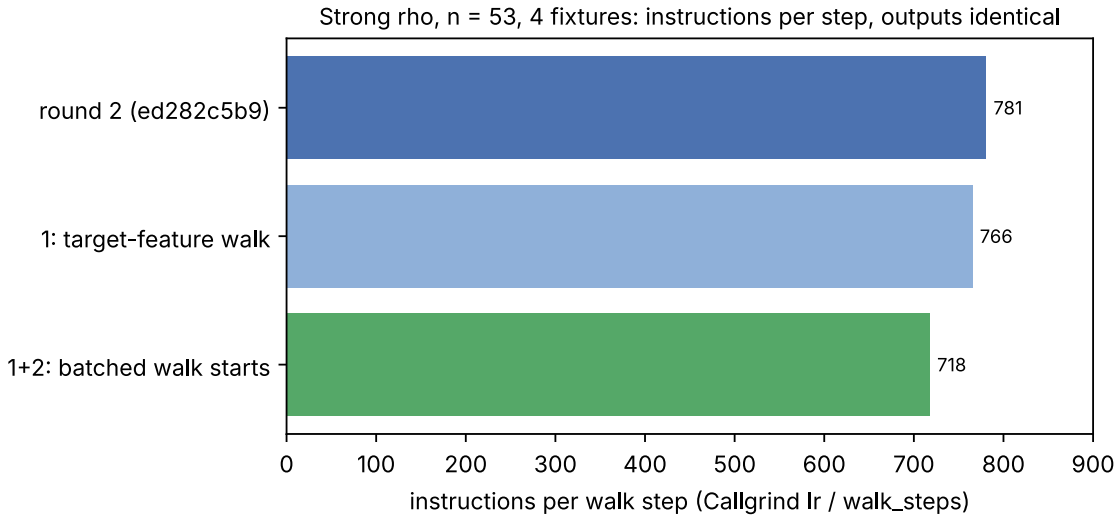


Figure 1: Instructions per walk step by arm (`runs/callgrind/`).

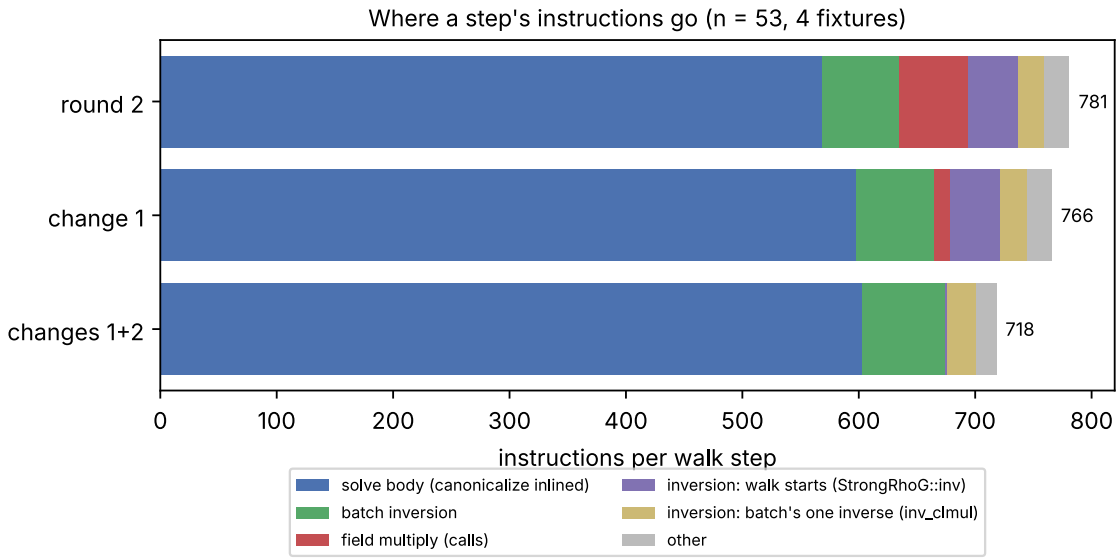


Figure 2: Where a step's instructions go, by arm.

1.3. Native time (measured)

A/A (round 2 × 6), then 5 interleaved A/B rounds, 32 processes under `isolated_bench`. Four were contended and are excluded: two A/A at n61 and two A/B round-2 runs at n53.

curve	round 2 ns/step	final ns/step	ratio	A/A spread, verdict
`icv1-f2m53-…`	106.8	105.9	1.008	5.6%: not established
`icv1-f2m61-…`	115.0	112.1	1.025	2.4% (4 runs): marginal, not claimed

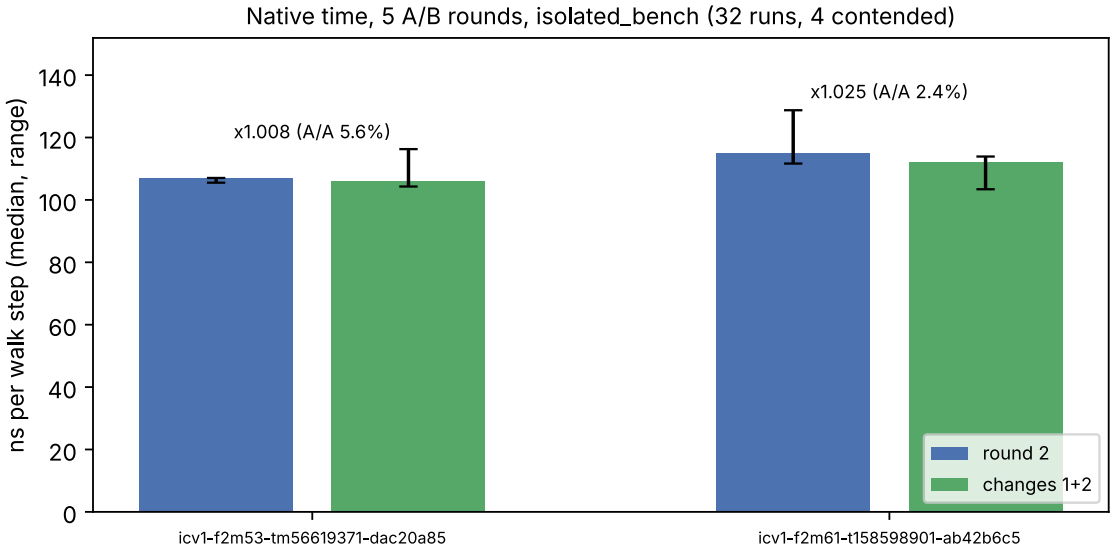


Figure 3: Native ns per walk step, median and range (`runs/time/`).

The instruction cut does not show as a clear time gain on this host. The removed calls and inversions were cheaper in time than in instructions.

IPC: the host has no hardware PMU, so no IPC is claimed.

1.4. Scope

- **What remains.** 718 instructions per step: about 600 in the `solve` body with canonicalization inlined (four basis conversions and the least-rotation search dominate), 71 in the walk's batched inversion, 25 in its single inverse. Going further would change the walk definition, which committed replays pin.

- **Counted units.** GAE records do not change. Future wall-time and instruction comparisons against this rho get harder for index calculus.
- **Not measured.** Hosts other than one x86-64 VM; the m = 83 wide instantiation, which gets change 2 only.